# TATA TECHNOLOGIES LTD.

**TECH PULSE FY-26 | AI/ML Capstone Project**

# PROJECT SYNOPSIS

## Student Details

Institute/University: Pimpri Chinchwad College of Engineering (PCCOE)
Programme / Batch: B.Tech Computer Engineering | Tech Pulse FY-26
Student Name: Atharv Rao
PRN: 123B1B024

## 1. Case Study ID and Project Title / Topic

Case Study ID (CS1 to CS5): CS1
Project Title / Topic: AUTOSAR Architecture Intelligence Assistant

## 2. Objectives

Define an evidence-backed AI/ML assistant for traceable AUTOSAR HLD analysis with synthetic/approved inputs, grounded citations, and human review.

### Objectives

- Convert HLD documents into structured, project-scoped architecture entities, relationships, and source evidence.
- Combine pgvector retrieval with bounded relational graph traversal for natural-language architecture analysis.
- Support provenance, confidence, explicit/inferred classification, human validation, audit history, impact analysis, and revision comparison.
- Provide a runnable web workspace with declared model/provider dependencies and evaluation evidence without exposing secrets or treating AI output as automatic truth.

## 3. Scope

Scope is limited to the submitted AUTOSAR HLD-analysis workflow, synthetic demonstration data, grounded answers, graph exploration, impact analysis, revision comparison, and human review.

### In Scope – To Be Implemented

- Ingest PDF, DOCX, XLSX, CSV, and Markdown with structural parsing and provenance preservation.
- Generate candidate AUTOSAR entities/relationships, resolve aliases, validate taxonomy, and persist evidence-linked relational graph facts.
- Provide hybrid retrieval (pgvector dense + PostgreSQL full-text, fused), bounded graph traversal, evidence sufficiency checks, cited and claim-verified answers, change-impact analysis, revision comparison, persisted 'potential inconsistency' findings, structured export (JSON/CSV/Excel) and review/audit workflows.
- Run locally or through Docker Compose using configurable LLM/embedding providers; use synthetic BrakeController and ABS HLD fixtures for the demonstration; process uploads through a sequential queue with a visible five-stage pipeline.

### Out of Scope – Will Not Be Implemented

- Confidential/customer HLDs, production enterprise identity integration, an approved malware-scanning policy, validation of OCR on real scanned HLDs, and large-scale private deployment.
- Claims of live-model or production retrieval accuracy: results on small synthetic fixtures are reported as indicators only.
- Automatic promotion of model output to trusted architectural knowledge without validation or review.

## 4. Proposed Approach

### Pre-trained model (name, version, local/private hosting)

Groq OpenAI-compatible API | openai/gpt-oss-120b | hosted external provider (synthetic data only); used for candidate extraction, ambiguous entity-merge decisions and answer drafting through LLMProvider. Local Ollama is supported by configuration.

### Embedding model

Gemini native embeddings API | gemini-embedding-2 | 768-dimensional output; accessed through EmbeddingProvider; model name and version are stored per chunk and a change requires an explicit re-embed.

### Vector store (FAISS / ChromaDB)

PostgreSQL 18 + pgvector (vector(768), HNSW cosine) as the system of record, plus PostgreSQL full-text search (tsvector + GIN) for the lexical channel; relational graph tables are used instead of FAISS/ChromaDB/Neo4j.

### RAG pipeline and/or AI-agent design – brief description

Upload queue -> validation/SHA-256 -> parser, section tree, structure-aware chunks (OCR if needed) -> embeddings (pgvector) -> LLM candidate extraction -> entity resolution (blocking, guards, rules, LLM only for ambiguous pairs, Union-Find) -> deterministic validation, confidence and findings -> explicit graph build with provenance. Query: planner -> dense + lexical retrieval and bounded graph traversal -> evidence bundle -> sufficiency check (two rounds at most) -> LLM drafts cited claims -> claims verified against the cited text -> grounded answer or 'insufficient evidence' -> human review (verify / edit / reject).

### Service layer and UI (e.g., FastAPI / Flask / Streamlit)

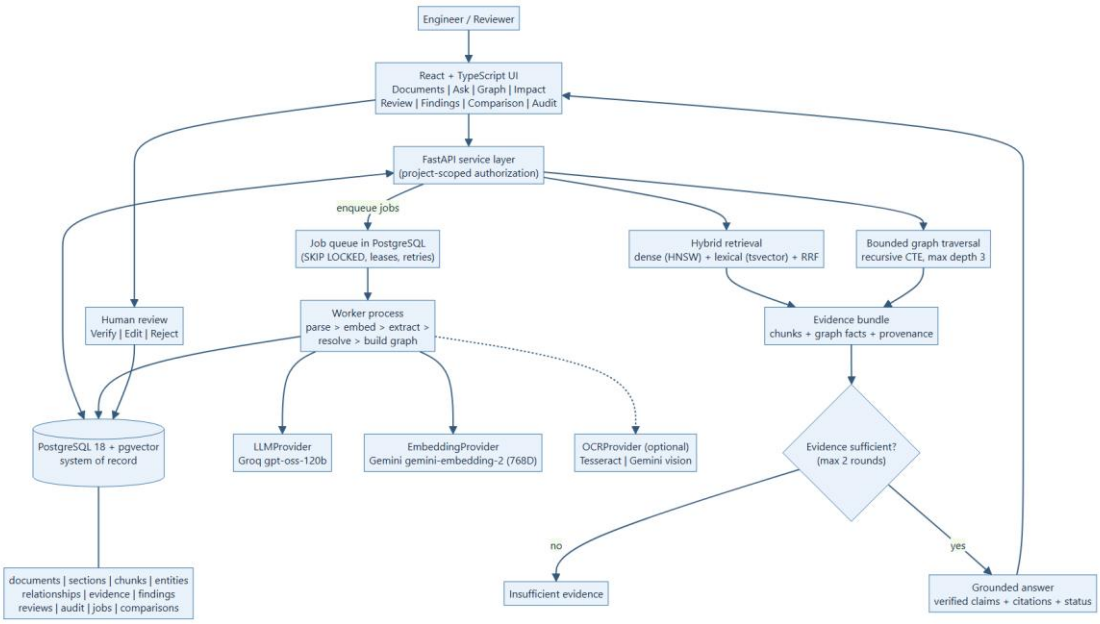
React 18 + TypeScript + Vite (3D graph with three.js); FastAPI + Pydantic v2 service layer; PostgreSQL-backed job queue with a separate worker; SQLAlchemy 2 + Alembic persistence; Docker Compose deployment.

### Tools / deterministic rules, if applicable

Pydantic schema validation, controlled relationship taxonomy, entity-resolution guards and rules, confidence as a pure function, project-scoped authorization, provenance validation, bounded recursive-CTE traversal, citation and claim verification, safe untrusted-source handling, retries/timeouts, audit log and review rules.

### Proposed System / Architecture

The system architecture is shown below; the editable Mermaid sources are Documentation/SYSTEM_ARCHITECTURE.mmd and Documentation/IMPLEMENTATION_WORKFLOW.mmd (rendered figures in Documentation/figures/).




## 5. Input Data / Knowledge Base

**Use approved or synthetic data only.**

### Data / Knowledge Sources

Synthetic AUTOSAR-style fixtures written for the project: BrakeController_HLD_v1.md and its revision BrakeController_HLD_v2.md, and a separate ABS_Subsystem_HLD.md (with a diagnostics table and a declared status), plus a labelled set of 23 questions for the retrieval evaluation. No confidential customer data.

### Approximate Size and Format

Small UTF-8 Markdown fixtures (about 1-2 KB each) representing components, interfaces, ports, signals, functions, dependencies, aliases, tables and revision changes; the pipeline also accepts PDF, DOCX, XLSX and CSV.

### Preprocessing and EDA Planned

Validate extension/MIME/size, hash and store source content (identical bytes rejected), clean headers/footers, build a section tree, create structure-aware chunks with an embedded breadcrumb, read declared document metadata, OCR scanned pages through a provider boundary, and record the embedding model/version per chunk. Numerical EDA is not applicable to the document fixtures.

## 6. Expected Outcomes and Evaluation Plan

### Expected Outputs – aligned to the case study

Runnable web workspace; project-scoped knowledge graph with provenance; cited natural-language analysis; review/audit workflow; impact analysis, findings and revision comparison; structured export; reproducible source, configuration and evaluation package with screenshots and PostgreSQL evidence.

### Evaluation Method – test question set / ground truth / metrics / validation approach

Retrieval and answer evaluation on a labelled set of 23 questions on the synthetic ABS HLD (19 answerable across components, aliases, interfaces, dependencies, ports, identifiers, timing, diagnostics and hierarchy; 4 unanswerable), run offline and with the live providers: Recall@k, MRR, citation validity, refusal accuracy, false refusals, dropped claims and latency. Manual review of end-to-end answers, graph facts, impact and revision results against a written ground truth, plus automated tests (392 backend, 29 frontend). Recorded live result: Recall@8 100%, MRR 0.97, citation validity 100%, refusal accuracy 4/4, false refusals 10.5% (2 of 19). Small, student-labelled synthetic sample: an indicator, not an accuracy claim.

### Responsible AI Measures – grounding and citations / privacy / security / human oversight

Source documents remain authoritative; model outputs are candidates (always PENDING until a human verifies, edits or rejects them, with the original kept); material claims require citations and are verified against them; absent facts are refused; explicit/inferred, confidence and validation state are always shown together; findings use 'Potential inconsistency detected'; synthetic data only with hosted models; project-scoped access, safe source handling, prompt-injection separation and audit.

### MLOps / Deployment Elements Planned – versioning / Docker / tracking

Versioned source, specifications and architecture decision records; 12 Alembic migrations with a drift test; Docker Compose; persistent job states with retries; audit log; versioned prompts and recorded embedding model/version; an offline and live evaluation script; a secret-scan script. No MLflow/model registry is claimed.

## 7. Planned AI-Tool Usage

Coding assistants used under student direction and the runtime models, with purpose and project artifact affected; this information is carried into the AI-tool usage declaration of the Technical Report.

### AI Tool / Model

Codex (OpenAI)

### Purpose

Specification-guided coding assistance, debugging, test development, documentation drafts, and submission packaging under student direction.

### Artifact Affected

Source code, tests, configuration documentation, technical documentation, evaluation/report artifacts, and submission structure.

### AI Tool / Model

Claude Code (Anthropic)

### Purpose

Later-phase coding assistance under student direction: review against the case study, pipeline, retrieval, entity-resolution, graph and UI implementation, tests, evaluation runs, evidence capture and updates of the report and submission package.

### Artifact Affected

Source code, tests, migrations, scripts, technical documentation, evaluation evidence, report and synopsis.

### AI Tool / Model

Groq / openai/gpt-oss-120b (runtime model)

### Purpose

Candidate extraction, ambiguous entity-merge decisions and grounded answer drafting through LLMProvider (output is validated, never trusted directly).

### Artifact Affected

Candidate graph facts and answer drafts.

### AI Tool / Model

Google Gemini / gemini-embedding-2 (runtime model)

### Purpose

Chunk and query embeddings through EmbeddingProvider (768 dimensions).

### Artifact Affected

Vector representations stored in pgvector.

## Student Declaration

☐ I will not change the approved case-study scope, major methodology, or implementation approach without prior faculty approval.

☐ I confirm that the work submitted under this synopsis represents my own understanding and contribution, and I will be responsible for explaining every significant part of the selected case study and its implementation.

Name and Signature: Atharv Rao  Date: 7 October 2026